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* If you cannot read my handwriting, I'm sorry. Please email me and I will resend you a type version based on my handwriting.

Problem 1

1a) 1. Make Query Resolution more accurate by providing accurate and complete answer after chatbot system understand user needs and intent.

2. Make user satisfaction higher by minimizing user effort and frustration after they get the speedy helps by chatbot system.

1b) Differences :

1. Scope : LLMs are a specific type of technology within NLP.

2. Technology : NLP covers big techniques, but LLMs are specifically deep learning models.

1c) Word embeddings convert words to be represented by dense numerical vector, make each words be mapped to a point in multi-dimensional space.

Word embeddings are used as a input layer for NLPs models.

1d) (i) Count-based embedding :

tool: CountVectorizer

Explain: For each document, where each dimension represents a words from vocabulary, create a vector, the value of that will be the raw count of the word.

(ii) Dense embedding :

tool: Word2Vec

Explain: Learn embedding by prediction, using shallow neural network, the word vectors are the learned weights of the network's hidden layer.

(iii) Pretrained embedding :

tool: BERT

Explain: Embeddings from large models pretrained on massive text corpora, it is generated based on the entire sentence.

1e) Tokenizer

f) Multiple layers can make the model build the more abstract and complex representations of the input text. The lower layers can get the basic syntax, and the relationships between the local words. The higher layers can understand the more complex semantics, the relationships in wider range, and the more abstract concepts.

1g) Decoder-only models, Chatbots are a text generation task. Decoder-only models' architecture can do it better use preceding tokens to predict the next token.

1h) Fine-tuning make a large pre-trained language model to train on a smaller domain specific dataset. It makes the general-purpose model to do better in specific task without the new model train.

1i) 1. BLEU: Bilingual Evaluation Understudy, focus on precision, measure how many n-grams in a sentence comparing response by chatbot and human. The higher score, the better fluency, the more similarity with human response.

ROUGE: Recall-Oriented Understudy Understudy for Gisting Evaluation, focus on recall, measure how many n-gram from human responses be captured by chatbot's response. It can evaluate it but include 100% of important key points.

Perplexity: It is a intrinsic metric, measure the ability that probability predicts a sample. NLP evaluates the feeling of surprised of model to a test model. The lower perplexity, the more confident and accurate of AI, the better language modeling ability.

Explanation of Problem 6

I added Batch Normalization at the end of each linear layer. So the layer inputs can be normalized, which fosters the stable training and the effective learn in deep network, so the loss can be reduced significantly.

Output :

Problem6:

Outputs for written part (Problem 6):

Epoch 1/20, Loss: 1378.572705

Epoch 2/20, Loss: 250.690187

Epoch 3/20, Loss: 229.055016

Epoch 4/20, Loss: 177.262192

Epoch 5/20, Loss: 161.983472

Epoch 6/20, Loss: 154.473780

Epoch 7/20, Loss: 151.038431

Epoch 8/20, Loss: 149.755361

Epoch 9/20, Loss: 153.608823

Epoch 10/20, Loss: 151.976125

Epoch 11/20, Loss: 148.925777

Epoch 12/20, Loss: 146.585869

Epoch 13/20, Loss: 147.592722

Epoch 14/20, Loss: 148.784853

Epoch 15/20, Loss: 149.124182

Epoch 16/20, Loss: 146.447904

Epoch 17/20, Loss: 145.771761

Epoch 18/20, Loss: 146.728699

Epoch 19/20, Loss: 145.836558

Epoch 20/20, Loss: 145.633695